

BOTTLENECKS

The visible pile is almost never the cause.

How to find the point that actually chokes the operation.

AND THE CONSEQUENCE

It is the effect of something that held the water back earlier.

THE CASCADE

Attacking the pile instead of the cause.



WHAT YOU SEE

You see work piling up at one stage...



THE MECHANISM

...you add more people or more hours there...



THE CONSEQUENCE

...and the pile reappears a bit further down, in three weeks.

High volume at one stage is usually a symptom, not the disease.

THE FORK

Stop counting items, look at the clock.

WATCH THE CLOCK

01

How many items are stuck? That is volume.

02

How long has the oldest one waited? That is the bottleneck.

03

Waiting on what and on whom? Here the problem gets a name.

The age of the queue reveals more than the size of the queue.

THE METHOD

The four choke points.

01 **THE TAP**

more work comes in than the operation can process

02 **THE NARROW PIPE**

one stage has less capacity than all the others

03 **THE VALVE**

one approval runs through one person, almost always you

04 **THE LEAK**

rework and errors send work backward, in silence

Each calls for a different action. Treating everything as capacity is the classic error.

IN PRACTICE, IN ORDER

The right action for each choke.

01 • THE TAP

The tap.

negotiate intake, prioritize at the door, say no with criteria

02 • THE NARROW
PIPE

The narrow pipe.

reinforce that stage only, growing the rest changes nothing

03 • THE VALVE

The valve.

transfer the ceiling and the judgment, do not approve faster

04 • THE LEAK

The leak.

measure rework, it is invisible and it eats the whole capacity

TWO MODELS

Chasing the symptom or finding the choke.

THE COMMON MODEL

Chasing the symptom

Looks where the pile is biggest

Adds people to the visible stage

Pushes the doers to go faster

The queue returns within weeks

THE DESIGNED MODEL

Finding the choke

Looks at the oldest stuck item

Asks waiting on what and on whom

Attacks intake, capacity or ceiling

The queue stops growing

MISTAKE AND CONSEQUENCE

Three things that spoil the diagnosis.

01 A team busy all the time.

- high occupancy hides waiting, everyone works and nothing ships.

02 Counting items instead of time.

- the queue can shrink in size and get worse in age.

03 Not measuring rework.

- the leak shows up in no report, but it consumes everything.

If the operation stops when one person is away, the bottleneck has a name and a badge.

THE CHAIN

The queue clock test.

01 List what is stuck and how long each item has waited.



02 Take the oldest one and ask, waiting on what?



03 The answer points at the real bottleneck, almost always a decision.



WHERE LEADERSHIP STEPS IN

Ten minutes with the queue on screen beat a two hour meeting.

RUN IT THIS WAY NEXT TIME

Half an hour to map the plumbing.



STEP 01

Write down where work piles up and who is waiting on what.

no judgment, just the map of the flow as it is today



STEP 02

Count how many things run through you for approval.

if it is more than five a week, the valve is you



STEP 03

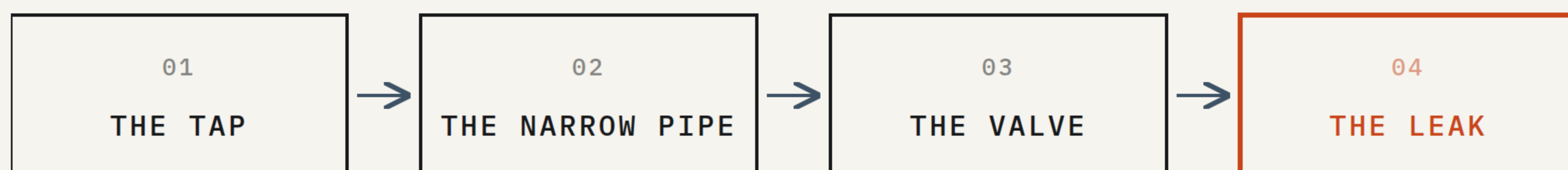
Measure how much work comes back from errors or changes.

measured rework is usually the biggest surprise of the month

Found the choke point? Attack that one only. Improving the rest changes no total.

THE MODEL THAT STICKS

The Four Choke Points, in the order that matters.



The bottleneck is not where the water stops. It is where someone is holding it back.

SAVE THIS FOR LATER

next time the operation feels slow.

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